

CNC Engineering Course - Table of Contents

CNC Engineering Course

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 3. **Thank You to AI** - Collaboration with AI Tools
 4. **License** - Course Terms and Conditions
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Course Modules

Foundation Modules

Module 01: Mechanical Frame and Structure

Module 02: Vertical Axis and Z-Stage

Module 03: Linear Motion Systems

Module 04: Control Electronics

Tool-Specific Modules

Module 05: Plasma Cutting

Module 06: Spindle and Rotary Tools

Module 07: Fiber Laser

Module 08: Waterjet Cutting

Advanced and Hybrid Systems

Module 09: Pick and Place Robot

Module 10: Robotic Arm

Module 11: Large-Format FDM 3D Printing

Module 12: Water-Jet Guided Laser Cutting

Control, Software, and Process Engineering

Module 13: EMI/EMC for Motion Control

Module 14: LinuxCNC HAL and Real-Time Control

Module 15: G-Code Standards and Post-Processing

Module 16: CAD Design for Manufacturable Parts

Module 17: Advanced Materials: Composites and Ceramics

Module 18: Industry 4.0 for CNC Manufacturing

Module 19: Advanced Control Systems

Module 20: Feeds and Speeds

Professional and Business Modules

Module 21: Metrology and Precision Measurement

Module 22: Quality Management Systems (QMS)

Module 23: Shop Organization and Management

Module 24: L.E.A.N. Strategies for CNC Manufacturing

Module 25: Work-Life Balance in CNC Manufacturing

Module 26: CNC Business Ownership and Management

Appendices

Appendix A: Material Properties

Appendix B: Hardware Specifications

Appendix C: Motor Drive Sizing

Appendix D: Linear Motion

Appendix E: Electrical Standards

Appendix F: G-code Reference

Appendix G: Safety Standards

Appendix H: Lubrication

Appendix I: Conversions

Appendix J: Troubleshooting

Appendix K: Vendors

Appendix L: Resources

Appendix M: Glossary

Appendix N: Contact

Appendix O: Updates

Appendix P: Mathematics

Appendix Q: QMS Templates

Appendix R: Organization Templates

Appendix S: LEAN Manufacturing Templates (2,910 lines)

Appendix T: Business Ownership Templates (6,612 lines)

Course Information

Total Modules: 26 comprehensive modules

Total Appendices: 20 reference appendices (including 9,500+ lines of templates)

Course Scope: Foundation through advanced topics and business ownership

Target Audience: Entry-level through experienced professionals and entrepreneurs

Course Version 1.0 / Last Updated: November 2025